

Robert Ashe

Applied AI Engineer · AI/Robotics Researcher · Co-Founder
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[LinkedIn](#) · [GitHub \(nthPerson\)](#) · [Project Portfolio](#)

Summary

Applied AI engineer and researcher building agentic systems and ML models that bridge software and the physical world. Co-author of BIM2RDT (an agentic AI safety-first framework that fuses building information models with LiDAR/IMU data into robot-ready construction-site digital twins) and co-founder of LARK Business Services, where I architected a multi-agent document-ingestion pipeline and RAG-grounded AI agents. Active research assistant on two SDSU labs (DiCE Lab; FAMAIL) and a Caltrans-funded traffic-safety study. Pursuing M.S. Big Data Analytics at SDSU (expected Spring 2027).

Education

San Diego State University – M.S. Big Data Analytics **Fall 2025 – Spring 2027 (expected)**
San Diego, CA

San Diego State University – B.S. Computer Science **2023 – 2025**
Major GPA 4.0 · Cumulative GPA 3.96 · Summa Cum Laude · Tau Sigma National Honor Society · 2025 Outstanding CS Graduate Award Nominee

San Diego Mesa College, A.S. Computer Science for Transfer | San Diego, CA **2023**
Valedictorian, High Honors

Experience

Research Assistant – SDSU DiCE Lab **2024 – Present**
San Diego State University · PI: Dr. Reza Akhavian

- Co-authored “BIM2RDT: An Agentic AI Safety-First Framework” (arXiv:2509.20705) an agentic-AI pipeline fusing BIM with live sensor data into robot-ready site digital twins for autonomous inspection, navigation, and safety monitoring.
- Lead developer on a Caltrans-funded study of outdoor advertising displays’ impact on driver distraction.
- Co-authored an i3ce 2025 paper on an accessible Python control framework for Unitree quadrupeds; integrated BIM and Unity simulation, plus an automated HAR data-labeling mechanism for construction-safety datasets.

Research Assistant – Fairness-Aware Multi-Agent Imitation Learning **Aug 2025 – Present**
San Diego State University · PI: Dr. Xin Zhang

- Developing a spatial-temporal data augmentation framework for taxi GPS trajectories to reduce bias amplification in behavioral models trained on human-generated data.
- Designing PyTorch augmentation operators that perturb spatial-temporal features while preserving task-relevant structure, yielding fairer multi-agent imitation-learning policies.
- Cut augmentation runtime from 16 days (CPU) to 39 minutes by partitioning workloads across CPU and GPU using HPC principles.

Co-Founder & Lead AI Engineer – LARK Business Services **Jan. 2026 – Present**
AI-native platform for merchant card processing professionals; co-founded with industry experts

- Architected a multi-agent document-ingestion pipeline that normalizes merchant card processing statements across processors and powers interchange optimization, downgrade detection, and effective-rate analysis.
- Built unstructured-data ingestion pipelines and RAG systems over interchange rules and processor knowledge to ground agent reasoning in authoritative sources.
- Built “Sidekick,” an AI agent for sales professionals (proposal prep, portfolio analytics, document Q&A); shipped multi-agent extraction and analysis engine, branded sales-proposal PDF generation, a portfolio analytics module, and agentic assistant.
- Built multi-tenant admin features (ISO/agent management, RBAC, user provisioning); leading integration with Echelon Payments.

Publications

- Co-authored, “*Building Information Models to Robot-Ready Site Digital Twins (BIM2RDT): An Agentic AI Safety-First Framework.*” arXiv:2509.20705 (2025). arxiv.org/abs/2509.20705
- Co-authored, “*Accessible Control Framework for Unitree Robots.*” Accepted, ASCE International Conference on Computing in Civil Engineering (i3ce) 2025.

Selected Projects

- **LARK Statement Analysis Engine** – Multi-agent ingestion + RAG over interchange rules; powers statement Q&A, downgrade detection, and effective-rate analysis for merchant card processors.
- **BIM2RDT Framework** – Agentic AI framework fusing BIM with real-time LiDAR/IMU data to produce safety-aware, robot-ready site digital twins (arXiv 2509.20705).
- **Car Sounds: DS-CNN Automotive Fault Diagnosis on Arduino Nano 33 BLE Sense Rev2** – Depthwise-separable CNN audio classifier on mel-spectrograms; PTQ-quantized and deployed to edge hardware for on-device automotive fault diagnosis. (M.S. coursework)
- **FEURcast – SPY Predictive Analytics Dashboard** – Gradient-boosting regressor predicting next-day SPY returns; end-to-end pipeline with walk-forward evaluation and an interactive dashboard. (M.S. coursework)
- **From Beats to Hits** – Model-selection study predicting Spotify track popularity from audio features; compared linear, tree-based, and gradient-boosted models. (M.S. coursework)

Technical Skills

- **Languages:** Python, JavaScript/TypeScript, C, C++, Java, C#, Scala, R, SQL, Bash
- **AI / ML:** PyTorch, TensorFlow, scikit-learn, gradient boosting, CNNs (depthwise-separable), imitation learning, RAG, multi-agent LLMs, PTQ, fairness/bias mitigation, statistical inference
- **Robotics & Built Environment:** BIM, Unity simulation, Unitree quadrupeds, LiDAR/IMU sensor fusion, HAR, DriveSafety RS-250 driving simulator, Pupil Core eye tracker, EMOTIV EPOC X 14 EEG
- **Data & Engineering:** Agentic software development, NumPy/Pandas/Matplotlib/Jupyter, SWITRS/geospatial crash-data analysis, Linux/Unix, Git/GitHub, Jira, multi-tenant SaaS, REST APIs, PDF generation
- **Hardware:** CAD design, production-scale 3D printing (300+ custom parts), edge-AI deployment on Arduino Nano 33 BLE Sense Rev2

Relevant Coursework

- **Graduate (M.S. Big Data Analytics, SDSU):** Machine Learning, Deep Learning, Big Data Analytics, Data Engineering, TinyML / Edge AI.
- **Undergraduate (B.S. Computer Science, SDSU):** Data Structures, Algorithms, Software Engineering, Machine Learning, Statistics, Programming in Python/C/C++/Java/JavaScript/Scala.